AI Assisted Coding (III Year) Assignment

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Lab 11 – Data Structures with AI: Implementing Fundamental Structures

Week 6 – Monday

Lab Objectives

- **Use AI to assist in designing and implementing fundamental data structures in Python**
- **Learn prompt-based design and optimization**
- **Improve understanding of Lists, Stacks, Queues, Linked Lists, Trees, Graphs, and Hash Tables**
- **Enhance code quality with comments and documentation**

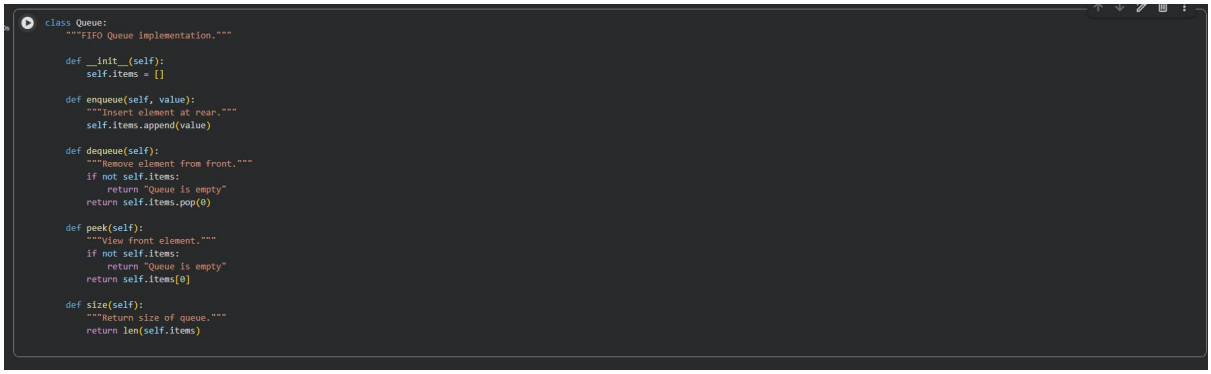
Task #1 – Stack Implementation

```python
class Stack:
    """Implementation of Stack using Python list."""

    def __init__(self):
        self.items = []

    def push(self, value):
        """Add an element to the top of the stack."""
        self.items.append(value)

    def pop(self):
        """Remove and return the top element."""
        if self.is_empty():
            return "Stack is empty"
        return self.items.pop()

    def peek(self):
        """Return the top element without removing it."""
        if self.is_empty():
            return "Stack is empty"
        return self.items[-1]

    def is_empty(self):
        """Check whether stack is empty."""
        return len(self.items) == 0
```

Explanation:

A stack is a linear data structure that follows the **LIFO (Last In First Out)** principle.
This means the element that is added last will be removed first.
In this implementation, a Python list is used to store elements.
The push() method adds elements to the end of the list, and the pop() method removes elements from the end.
This stack is useful in real-world scenarios such as undo/redo operations and function call management.

Task #2 – Queue Implementation



```python
class Queue:
    """FIFO Queue implementation."""

    def __init__(self):
        self.items = []

    def enqueue(self, value):
        """Insert element at rear."""
        self.items.append(value)

    def dequeue(self):
        """Remove element from front."""
        if not self.items:
            return "Queue is empty"
        return self.items.pop(0)

    def peek(self):
        """View front element."""
        if not self.items:
            return "Queue is empty"
        return self.items[0]

    def size(self):
        """Return size of queue."""
        return len(self.items)
```

Explanation:

A queue follows the **FIFO (First In First Out)** principle.
The element added first is removed first.
In this implementation, elements are added at the end using enqueue() and removed from the front using dequeue().
Queues are commonly used in real-life systems like ticket booking, CPU scheduling, and order processing.

Task #3 – Singly Linked List

```python
class Node:
    """Node of a singly linked list."""

    def __init__(self, data):
        self.data = data
        self.next = None


class LinkedList:
    """Singly linked list implementation."""

    def __init__(self):
        self.head = None

    def insert(self, value):
        """Insert at end."""
        new_node = Node(value)
        if not self.head:
            self.head = new_node
            return

        temp = self.head
        while temp.next:
            temp = temp.next
        temp.next = new_node

    def display(self):
        """Display linked list."""
        temp = self.head
        while temp:
            print(temp.data, end=" -> ")
            temp = temp.next
        print("None")
```

Explanation:

A linked list is a dynamic data structure where elements are stored in nodes.
Each node contains data and a reference to the next node.
Unlike arrays, linked lists do not require continuous memory.
Insertion is easier compared to arrays, especially when frequent updates are needed.
Linked lists are used in music playlists, navigation systems, and memory management.

Task #4 – Binary Search Tree

```
class BST:
    """Binary Search Tree implementation."""

    def __init__(self, value=None):
        self.value = value
        self.left = None
        self.right = None

    def insert(self, value):
        """Recursive insert."""
        if self.value is None:
            self.value = value
        elif value < self.value:
            if self.left:
                self.left.insert(value)
            else:
                self.left = BST(value)
        else:
            if self.right:
                self.right.insert(value)
            else:
                self.right = BST(value)

    def inorder(self):
        """In-order traversal."""
        if self.left:
            self.left.inorder()
        print(self.value, end=" ")
        if self.right:
            self.right.inorder()
```

Explanation:

A Binary Search Tree stores data in a hierarchical structure.
The left child always contains smaller values and the right child contains larger values.
In-order traversal prints elements in sorted order.
BSTs are useful for searching, sorting, and implementing databases and file systems.

Task #5 – Hash Table with Chaining

```
class HashTable:
    """Hash table using chaining."""

    def __init__(self, size=10):
        self.size = size
        self.table = [[] for _ in range(size)]

    def _hash(self, key):
        return hash(key) % self.size

    def insert(self, key, value):
        """Insert key-value."""
        index = self._hash(key)
        for pair in self.table[index]:
            if pair[0] == key:
                pair[1] = value
                return
        self.table[index].append([key, value])

    def search(self, key):
        """Search value by key."""
        index = self._hash(key)
        for pair in self.table[index]:
            if pair[0] == key:
                return pair[1]
        return None

    def delete(self, key):
        """Delete key."""
        index = self._hash(key)
        for i, pair in enumerate(self.table[index]):
            if pair[0] == key:
                self.table[index].pop(i)
                return
```

Explanation

A hash table stores data in key-value pairs.
It provides very fast searching because keys are converted into indexes using a hash function.

When collisions occur, chaining is used to store multiple values at the same index.
Hash tables are widely used in databases, dictionaries, and caching systems.

Task #6 – Graph (Adjacency List)

```
class Graph:
    """Graph using adjacency list."""

    def __init__(self):
        self.graph = {}

    def add_vertex(self, vertex):
        if vertex not in self.graph:
            self.graph[vertex] = []

    def add_edge(self, v1, v2):
        self.graph.setdefault(v1, []).append(v2)

    def display(self):
        for v in self.graph:
            print(v, "->", self.graph[v])
```

Explanation:

A graph represents relationships between nodes.
Using an adjacency list is memory efficient.
Graphs are used in maps, social networks, routing algorithms, and recommendation systems.

Task #7 – Priority Queue

```
import heapq


class PriorityQueue:
    """Priority Queue using heap."""

    def __init__(self):
        self.heap = []

    def enqueue(self, value, priority):
        """Lower number = higher priority."""
        heapq.heappush(self.heap, (priority, value))

    def dequeue(self):
        if not self.heap:
            return "Empty"
        return heapq.heappop(self.heap)[1]

    def display(self):
        print(self.heap)
```

Explanation:

A priority queue processes elements based on priority rather than insertion order.
The heapq module maintains the heap property efficiently.
Priority queues are used in CPU scheduling, emergency services, and shortest path algorithms.

Task #8 – Deque

```
from collections import deque


class DequeDS:
    """Double-ended queue."""

    def __init__(self):
        self.dq = deque()

    def insert_front(self, value):
        self.dq.appendleft(value)

    def insert_rear(self, value):
        self.dq.append(value)

    def remove_front(self):
        if not self.dq:
            return "Empty"
        return self.dq.popleft()

    def remove_rear(self):
        if not self.dq:
            return "Empty"
        return self.dq.pop()
```

Explanation:

A deque allows flexible insertion and deletion from both ends.
It is more efficient than lists for such operations.
Deques are used in sliding window problems and task scheduling.

Task #9 – Campus Resource Management System

Feature Mapping Table

Feature	Data Structure	Justification
Student Attendance	Deque	Students enter and exit from both ends, making deque efficient for real-time logging.
Event Registration	Hash Table	Fast search, insertion, and deletion based on student ID.
Library Borrowing	Linked List	Dynamic insertion and removal of books without shifting elements.
Bus Scheduling	Graph	Routes and stops form a network structure.
Cafeteria Orders	Queue	Orders processed in arrival order (FIFO).

Implementation – Cafeteria Order Queue

```
class CafeteriaQueue:
    """Queue system for cafeteria orders."""

    def __init__(self):
        self.orders = []

    def place_order(self, student):
        """Add new order."""
        self.orders.append(student)

    def serve_order(self):
        """Serve next order."""
        if not self.orders:
            return "No orders"
        return self.orders.pop(0)

    def display(self):
        print("Pending:", self.orders)


# Example
cq = CafeteriaQueue()
cq.place_order("Akshay")
cq.place_order("Rithwik")
cq.display()
print("Served:", cq.serve_order())
```

```
Pending: ['Akshay', 'Rithwik']
Served: Akshay
```

Explanation:

In the Campus Resource Management System, different features require different data structures based on how data is used. Student attendance tracking needs to record both entry and exit of students, so a deque is suitable because it allows insertion and removal from both ends. The event registration system requires fast searching and deletion of participant details, which can be efficiently handled using a hash table. For library book borrowing, books need to be managed in an ordered manner based on due dates, so a binary search tree is useful as it maintains sorted data. Bus scheduling involves routes and stops that are interconnected, making a graph the best choice to represent these relationships. The cafeteria order system serves students in the order they arrive, so a queue is ideal because it follows the FIFO principle. Choosing these data structures improves efficiency and performance of the campus system.

Task #10 – Smart E-Commerce Platform

Feature Mapping Table

Feature	Data Structure	Justification
Shopping Cart	Linked List	Dynamic addition and removal without shifting.
Order Processing	Queue	Orders handled sequentially.
Top-Selling Products	Priority Queue	Highest sales given priority.
Product Search	Hash Table	Fast lookup by product ID.
Delivery Planning	Graph	Represents routes between locations.

Implementation – ProductSearch using Hash Table

Feature	Data Structure	Justification

```python
class ProductSearch:
    """Fast lookup of products."""

    def __init__(self):
        self.products = {}

    def add_product(self, product_id, name):
        """Add product."""
        self.products[product_id] = name

    def search_product(self, product_id):
        """Search by ID."""
        return self.products.get(product_id, "Not found")

    def remove_product(self, product_id):
        """Delete product."""
        if product_id in self.products:
            del self.products[product_id]


# Example
ps = ProductSearch()
ps.add_product(101, "Laptop")
ps.add_product(102, "Mobile")
print(ps.search_product(101))
```

```
Laptop
```

Conclusion

This lab demonstrated how AI can assist in:

- **Designing and implementing core data structures**
- **Improving documentation and readability**
- **Selecting optimal structures for real-world applications**
- **Enhancing efficiency and scalability**